STOLEN STRATA: Brick Kilns and the Loss of Karewa Tableland

Budgam, Kashmir, 1993–2025, with a correction to my earlier estimate

Executive Summary · Revised 7 October 2026 · Sakshi D. Maske

The earlier summary and preprint (4 September 2026) reported a rise in bare-earth share from 1.84% to 8.43% and a net increase of 190.3 ha. Those numbers are withdrawn. This summary replaces them.

What changed, in one paragraph

My first version said bare ground on Kashmir's karewa terraces had more than quadrupled since 1994, almost all of it after 2015. That jump sat exactly where the analysis switched from Landsat to Sentinel-2. When I ran the same polygons, the same rule and the same statistic on Landsat alone, the 2025 figure came out at 0.07%, not 8.43%. The rise came from mixing two differently built satellite products, not from the ground. So I rebuilt the study: a new terrace map checked against a published geological map, a time series that never compares one sensor with another, and an accuracy sample I labelled by hand. What remains is smaller.

The questions

1. How much karewa tableland that was vegetated in 2013–2015 was no longer vegetated in 2023–2025, and does it convert faster than comparable land?
2. What did that land become, and how much of it now lies inside brick-kiln fields?
3. Where and when did the change happen?

Method in one line: terraces mapped from the Copernicus elevation model (slope, height above drainage, scarp share); every 30 m pixel's yearly peak NDVI followed in Landsat 8/9 alone; a strict and a looser change test; two stratified random samples, 240 points in all, labelled by hand to say what the flagged land is now and what it was in 2013–2015.

What the study now shows

Study box	About 2,160 km² across Budgam, Pulwama and Srinagar districts
Terraces mapped	180 scarp-bounded karewa tablelands, 173.3 km²
Earlier headline	1.84% → 8.43%, +190.3 ha: **artefact of mixing two satellite products, withdrawn**
Same rule and statistic, Landsat only	2.02% (1994), 7.31% (2005, a year of too few cloud-free summer images), 2.16% (2015), 0.07% (2025)
Vegetated in 2013–15, gone by 2023–25, on terraces	107 ha by a strict test, 329 ha by a looser one
Of the flagged land, not vegetated today	**about 299 ha of 331 (roughly 285–313 ha)**

Vegetated in 2013–15 imagery and not vegetated today	**about 262 ha (roughly 243–281 ha)**
Of the flagged land, inside brick-kiln fields	**about 210 ha (roughly 189–231 ha)**, under one written rule for where a kiln field ends; 204 ha by the first sample, 217 ha by the second
Flagged sample points that show vegetation in 2013–15 imagery	158 of 180 (59 of 60 strict, 99 of 120 looser)
Kiln-field land where the older image also shows vegetation	about 197 ha (roughly 176–218 ha)
Net loss in the two kiln belts since the mid-1990s	roughly 407 ha, indicative only: the record before 2013 is thin
The other 140 km^2 of terraces	Some conversion after 2013, at the same rate as comparable flat land

Where and when

Rangeen Kultreh (Chadoora tehsil). Green in every usable year from 1993 to 2016 (one anomalous year, 2001, aside). Then a kiln field opens: 24 ha bare in 2017, 75 ha in 2018, and today about a quarter of those terraces. Landsat 7 and Landsat 8 agree on the change; Sentinel-2 agrees on the present state. A Pollution Control Committee report to the National Green Tribunal says one of the kilns there was commissioned in 2017 without consent and ordered closed in 2018, with 20 kilns within a kilometre. I got the 2017 date from the satellite record before I found that document.

Bandagam–Batapora tablelands. Older, and also the largest recent loss. Land that never greens up was about 100 ha in the mid-1990s, about 300 ha by 2008–2012 and about 450 ha now; the early part rests on few images. Inside the reliable Landsat 8/9 record (2013–2025) the looser test finds 142 ha of loss here, more than at Rangeen Kultreh, spread over many patches. The Pollution Control Committee's list of October 2023 has registered kilns under most of the village names in this belt, without locations.

Everywhere else. Of 180 terraces, 27 lost a hectare or more after 2013 by the looser test and only 8 by the strict one. Taken as a whole, terraces convert several times faster than other flat, raised land, but that difference comes entirely from the two belts: outside them, and compared polygon by polygon, terraces convert at the same rate as comparable land.

How I checked it

- **One sensor family at a time.** Landsat 8/9 for the main test, Landsat 7 on its own as a second opinion on the change (it confirms 86% of the pixels), Sentinel-2 (from 2019) on the present state.
- **Terrace map against geology.** In the original (eastern) study box, 80% of the mapped terrace area lies on Karewa formations on a published map. The western strip, including the Bandagam–Batapora belt, has no geological map.
- **My own choices.** Nine versions of the terrace rule and 51 versions of the change thresholds. The hectares move; the contrast and the locations do not.

- **By eye.** Two random samples, 240 points in all, labelled without knowing which group each came from, against the most recent imagery available. The vegetation is gone at 100% of strict detections and 86% of looser ones (one of the 120 looser points could not be told). Almost all strict detections (93%) lie inside a kiln field; about half of the looser ones do, the rest being houses, roads and other bare ground. At first the two samples disagreed on how much of the land is kiln land (141 against 232 ha), because they had counted bare ground, tracks and cleared plots at the edge of a kiln field differently. I wrote down one rule for where a kiln field ends and relabelled the 86 edge points of both samples under it; the samples then agree (204 and 217 ha).
- **Back in time, point by point.** I looked up the 180 flagged points in Google Earth imagery of 2013–2015. Vegetation is visible at 158 of them: 59 of the 60 strict points and 99 of the 120 looser ones. Twenty-one were not green in that image (kiln ground or tracks already there, buildings, yards and roads, bare plots) and one could not be told. Seven of the twenty-one are points I count as kiln field today, so the kiln figure restricted to land the older image shows as vegetated is about 197 ha instead of 210.

What I could not show

- **How deep the ground was dug.** Every free elevation model is older than the Rangeen Kultreh kiln field, and the space laser data has only nine shots on it.
- **A saffron link.** The saffron tablelands at Pampore are about 7 km or more from the main kiln field and show no conversion of this kind in 2013–2025. The saffron value-at-risk figure in the earlier version is withdrawn.
- **The whole karewa.** I map scarp-bounded tablelands, about a quarter of the Karewa formation area on the geological map. The terrace map itself has no accuracy figure yet.
- **An independent kiln figure.** The kiln area rests on one written rule applied by one labeller, mostly from the setting of each point within about 100 m of bricks, pits or a kiln. Kiln-field outlines on dated imagery, or a second labeller, would test it.
- **A fully blind check of the past.** In the first sample the older imagery was looked up only for points already flagged; the second sample was labelled blind. Mostly one image of September 2014 was used, and "vegetated" there includes rough grass and scattered trees. One image shows one day; the test uses the peak of the year.
- **Roads, settlements and the 25 "degraded terraces".** Those analyses depended on the withdrawn layer and are dropped.

Corrected in October 2026, and what is still open

- The long 1993–2025 series first applied the Landsat 8/9 adjustment in the reverse direction. It was recomputed with the right coefficients in October 2026; the three-decade figure for the two belts rose from about 335 ha to about 407 ha. The 2013–2025 results never used that adjustment.
- Still open: the Landsat values in the check of the earlier estimate (the 0.07% for 2025) come from the original-box series, which still carries the reversed adjustment and has not yet been recomputed.

Why it matters

If the karewas are to be protected, this study points to a short and specific list of places, and one of them was vegetated land ten years ago. A satellite claim like my first one should also be tested on a single sensor before it is published; mine was not.

Where things are

Paper: `SS_Research_Paper.md`. Scripts, tables, terrace layers and the labelled sample: `v2_redesign/`. Day-by-day record: `SS_Development_Log.md`.